

10.3 Practical biology and investigative skills

Teachers should make the practical and investigative skills criteria available to students to enable them to understand what is expected of them.

Students should further develop their practical skills, whichever approach (concept or context) has been taken. Students should carry out the recommended core practicals in Units 4 and 5.

Students should carry out practical work during the IA2 course, which should be verified by the teacher using the criteria below.

Other practical-related skills, including analysis and evaluation of data may be assessed in the externally assessed components for Units 4 and 5.

The teacher should verify students' ability to:

- a Use apparatus skilfully and safely
 - i Apparatus and materials are handled correctly and safely and manipulative techniques are used in an appropriate and safe manner.
 - ii The practical work is carried out in an organised, methodical and safe manner, with due consideration of the wellbeing of living organisms and the environment.
- b Produce and record reliable and valid results
 - i Measurements and observations are made with precision and recorded in a structured manner; variables are identified and the validity and reliability of results are justified.
 - ii Possible systematic errors and random errors in generating results are identified and explained.
- c Present and analyse data
 - i Use appropriate methods to analyse results, present data and identify trends, patterns and/or observations.
 - ii Any apparent anomalies and inconsistencies are described, the methodology is evaluated and suggestions are made to improve or further the work of the investigation.

Students are expected to know the conventions for the collection and presentation of data set out in the document issued by the Institute of Biology:

Biological Nomenclature – Standard Terms and Expressions Used in the Teaching of Biology, 3rd Edition (Institute of Biology, 2000) ISBN 0900490365, or visit: www.iob.org.